

assignment circular2 - good for test2

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:36pm

Quiz Instructions

tutorial available



Question 1 9.1 pts

A Ferris wheel has radius 5.0 m and makes one revolution every 8.0 s with uniform rotation. A person who normally weighs 670 N is sitting on one of the benches attached at the rim of the wheel. What is the apparent weight (the normal force exerted on her by the bench) of the person as she passes through the highest point of her motion? about

☐

350N

☐

670N

☐

880N

☐

460N



Question 2 9.1 pts

A 600-kg car is going around a banked curve with a radius of 110 m at a steady speed of 24.5 m/s. What is the appropriate banking angle so that the car stays on its path without the assistance of friction?

☐

60.9°

☐

56.2°

☐

33.8°

☐

29.1°

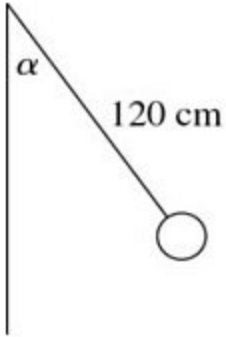
☐

13.5°



Question 3 9.1 pts

A 20-g bead is attached to a light 120-cm-long string as shown in the figure. This bead moves in a horizontal circle with a constant speed of 1.5 m/s. What is the tension force that the string exerts on the bead if the angle α is measured to be 25° ?



0.22 N



0.20 N



0.089 N



0.041 N



0.46 N



Question 4 9.1 pts

A car moving at a steady 10 m/s on a level highway encounters a bump that has a circular cross-section with a radius of 30 m. The car maintains its speed over the bump. What is the force exerted by the seat of the car on a 60.0-kg passenger when the car is at the top of the bump? this is a tricky question. You are going over a bump. Its similar to a passenger riding a ferris wheel and being at the top of the wheel. 2 forces are acting on you. The normal and the weight. one is larger. (in - out = ma or $mg - N = ma$)



390 N



790 N



490 N



200 N



590 N



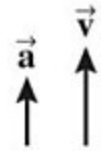
Question 5 9.1 pts

Shown below are the velocity and acceleration vectors for an object in several different types of motion. In which case is the object's velocity changing while its speed is not changing? (speed is the magnitude of the velocity.) Enter a upper case letter.

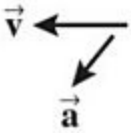
A)



B)



C)



D)



1.



Question 6 9.1 pts

When a car goes around a banked circular curve at the proper speed for the banking angle, what force cause it to follow the circular path?



No force causes the car to do this because the car is traveling at constant speed and therefore has no acceleration.



gravity



the friction force from the road



the normal force from the road



Question 7 9.1 pts

When an object moves in uniform circular motion, the direction of its acceleration is



is directed toward the center of its circular path.



depends on the speed of the object.



is directed away from the center of its circular path.



in the opposite direction of its velocity vector.



in the same direction as its velocity vector.



Question 8 9.1 pts

A particularly scary roller coaster contains a loop-the-loop in which the car and rider are completely upside down. If the radius of the loop is 13.2 m, with what minimum speed must the car traverse the loop so that the rider does not fall out while upside down at the top? Assume the rider is not strapped to the car.



11.4 m/s



14.9 m/s



12.5 m/s



10.1 m/s



Question 9 9.1 pts

A 0.50-kg toy is attached to the end of a 1.0-m very light string. The toy is whirled in a horizontal circular path on a frictionless tabletop. If the maximum tension force that the string can exert on an object without breaking is 350 N, what is the maximum speed the toy can have without breaking the string?



26 m/s



13 m/s



19 m/s



700 m/s



Question 10 9.1 pts

A jet plane flying 600 m/s experiences an acceleration of 4.0 g when pulling out of the circular section of a dive. What is the radius of curvature of this section of the dive?



0.64 km



1.2 km



7.0 km



5.8 km



9.2 km



Question 11 9.1 pts

A 1000-kg car is moving at 30 m/s around a horizontal unbanked curve whose diameter is 0.20 km. What is the magnitude of the friction force required to keep the car from sliding?



900 N



3000 N



9000 N



300 N



9800 N

Not saved

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